INCH-POUND

MIL-DTL-25026C
16 May 2011
SUPERSEDING
MIL-I-25026B
15 February 1985

DETAIL SPECIFICATION

INDICATOR, TEMPERATURE, THERMOCOUPLE, ENGINE EXHAUST, TYPE MJ-4

Inactive for new design after 24 March 1999.

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of single pointer, engine exhaust thermocouple temperature indicator designated Type MJ-4.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

Comments, suggestions, or questions on this document should be addressed to Defense Logistics Agency - Aviation, ATTN: DLA Aviation-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil.

AMSC N/A FSC 6685

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:20Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-7071	- Indicator, Temperature, Thermocouple, Hermetically Sealed, General Specification for
MS33639	- Case, Instrument, Clamp-Mounted, Aircraft

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property

(Copies of these documents are available online at https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY

ANSI/ASQ Z1.4	- Sampling Procedures and Tables by Inspection of Attributes

(Copies of ANSI/ASQ Z1.4 are available online at www.asq.org or from ASQ, 600 North Plankinton Avenue, Milwaukee, Wisconsin 53203)

AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA

NASM961	- Washer, Flat, Electrical
NASM35214	- Screw, Machine-Pan Head, Cross-Recessed, Brass, UNC-2A
NASM35215	- Screw, Machine-Pan Head, Cross-Recessed, Brass, UNF-2A
NASM35338	- Washer, Lock-Spring, Helical, Regular (Medium) Series

(Copies of these documents are available online at www.aia-aerospace.org/or from Aerospace Industries Association of America, 1250 I St. NW, Suite 1100, Washington DC, 20005)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:20Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 General. The requirements specified in MIL-I-7071, except qualification requirements, shall be applicable as requirements of this specification. Additional requirements shall be specified herein.

3.2.1 Precedence. In the event the requirements of MIL-I-7071 and this specification conflict, the requirements of this specification shall govern.

3.3 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5 Construction.

3.5.1 Case. The indicator shall conform to figures 1 and 2.

3.5.2 Case back. The back of the indicator case shall be distinctly marked to indicate "+" for the chromel terminal stud and "-" for the alumel terminal stud. The statement "USE 8 OHM C-A LEADS" shall be plainly marked on the back of the case.

3.5.3 Dial. The dial of the indicator shall conform to figure 1.

3.5.3.1 Scale arc. The indicator shall provide a minimum scale arc angle of 250 degrees. The length of the scale arc shall not be less than 6.25 inches as measured on the outer arc of the scale. The mechanism shall not be geared to obtain the specified scale arc.

3.5.3.2 Scale distribution. The scale of the indicator shall be as linear as is practical and shall not be contracted by more than 20 percent between any two adjacent scale graduations.

3.5.4 Pointer. The shape of the pointer shall facilitate reading of any point in the range of the dial.

3.5.4.1 Pointer overlap. The pointer length shall be of such that the pointer tip will extend into the longest scale graduation a distance of 0.094 ± 0.016 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:20Z
Check the source to verify that this is the current version before use.

3.5.4.2 Pointer overswing. The pointer overswing shall not be less than 3 angular degrees beyond each end of the scale.

3.6 Weight. The weight of the complete indicator shall not exceed 1.2 pounds.

3.7 Identification of products. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

4. VERIFICATION

4.1 General specification. The indicator shall be subjected to the verification provisions in accordance with MIL-I-7071 except as specified herein.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection – see MIL-I-7071 qualification testing
b. Conformance inspection – see MIL-I-7071 quality conformance testing

4.2 Test conditions.

4.2.1 Test tolerance. The tolerances specified in Table I, not enclosed in parentheses, are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in Table I, enclosed in parentheses, are applicable only when the indicators are subjected to a specific test independent of sequence.

4.2.2 Sampling. Lot formation, sampling, and retest shall be in accordance with ANSI/ASQ Z1.4.

4.2.2.1 Sampling plan A. Sampling plan A shall be inspection level II.

4.2.2.2 Sampling plan B. Sampling plan B shall be inspection level S-1.

4.3 Test methods. Unless otherwise specified, the test listings below are only the tolerances or requirements of the tests specified in MIL-I-7071. Lack of listing herein does not waive the test requirement.

4.3.1 Individual tests. Testing shall be in accordance with MIL-I-7071 individual tests.

4.3.1.1 Scale error at room temperature. The indicator pointer shall first be set to read 700°C when the millivoltage for 700°C, as specified in the Electromotive force for Chromel-Alumel thermocouple table of MIL-I-7071, for the ambient room temperature is applied to the terminals of the indicator through the specific resistance in series with the indicator. The test shall then be conducted at the specified test points of table I herein. The scale errors shall not

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:20Z
Check the source to verify that this is the current version before use.

exceed the tolerance specified. If greater overall indicator accuracy can be obtained by setting the pointer to 700±5°C with the specified voltage for 700°C, the pointer shall be so positioned.

TABLE I – Test Tolerances.

Test Point (°C)	Scale Error at Room Temp (± °C)	Scale Error at Low Temp (± °C)	Scale Error at High Temp (± °C)	Scale Error at Room Temp After Vibration (± °C)
0	60 (60)	75 (75)	70 (70)	75 (70)
200	40 (40)	55 (55)	50 (50)	55 (50)
400	30 (30)	45 (45)	40 (40)	45 (40)
500	20 (20)	35 (35)	30 (30)	35 (30)
600	10 (10)	25 (25)	20 (20)	25 (20)
700	5 (5)	20 (20)	15 (15)	20 (15)
800	10 (10)	25 (25)	20 (20)	25 (20)
900	20 (20)	35 (35)	30 (30)	35 (30)
1000	20 (20)	35 (35)	30 (30)	35 (30)

Test Point (°C)	Extreme Low Temp Exposure		Scale Error at Room Temp After High Temp (± °C)	Scale Error at Room Temp After Salt Spray (± °C)
	Scale Error at Low Temp (± °C)	Scale Error at Room Temp After Low Temp (± °C)		
0	90 (75)	80 (65)	80 (65)	80 (60)
200	70 (55)	60 (45)	60 (45)	60 (40)
400	60 (45)	50 (35)	50 (35)	50 (30)
500	50 (35)	40 (25)	40 (25)	40 (20)
600	40 (25)	30 (15)	30 (15)	30 (10)
700	35 (20)	25 (10)	25 (10)	25 (5)
800	40 (25)	30 (15)	30 (15)	30 (10)
900	50 (35)	40 (25)	40 (25)	40 (20)
1000	50 (35)	40 (25)	40 (25)	40 (20)

4.3.1.2 Position error. The position errors of the indicator shall be conducted at any point on the scale between 400°C and 900°C by applying the test voltage and the maximum change in indication shall not exceed 25°C.

4.3.1.3 Friction error. The maximum difference in reading, before and after tapping, shall not exceed 30°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:20Z
Check the source to verify that this is the current version before use.

4.3.1.4 Zero adjustment. Power shall be applied to produce a 700°C reading. When the indicator zero adjustment is operated throughout its range, the resulting pointer motion shall not be less than 40°C above and below the 700°C reading. The total motion of the pointer shall not exceed 200°C. The pointer shall be reset to the initial reading with the initial power applied.

4.3.1.5 Insulation resistance. The insulation resistance shall be a minimum of 20 megohms.

4.3.2 Sampling plan A tests. Testing shall be accordance with MIL-I-7071 sampling plan A tests.

4.3.2.1 Scale error at low temperature. The scale error shall not exceed the tolerances specified in Table I when subjected to the test requirements of MIL-I-7071. If there is a tendency for the pointer to the stick to the pointer stops, a light tap on the indicator shall be sufficient to free the pointer.

4.3.2.2 Scale error at high temperature. The scale error shall not exceed the tolerances specified in Table I when subjected to the test requirements of MIL-I-7071. If there is a tendency for the pointer to the stick to the pointer stops, a light tap on the indicator shall be sufficient to free the pointer.

4.3.2.3 Magnetic effect. When the indicator is subjected to the magnetic effect test of MIL-I-7071, the magnetic compass shall not deflect more than 5 degrees.

4.3.2.4 Vibration error. The indicators shall be subjected to the test requirements of MIL-I-7071. The maximum amplitude of pointer oscillation throughout the frequency range shall not exceed a range of 20°C within the temperature range of 400°C to 900°C and shall not exceed 30°C for the remainder of the scale. The maximum difference between the original position of the pointer before vibration and the position of the pointer during vibration shall not exceed 15°C between 400°C and 900°C and shall not exceed 25°C for the remainder of the scale. Pointer oscillation shall not be considered as pointer variation. The diameter of the circular motion of vibration shall be 0.009 inches to 0.011 inches

4.3.2.5 Vibration failure. The scale errors at room temperature after this test, as specified of MIL-I-7071, shall not exceed the tolerances specified in Table I. The zero adjustment shall not be reset after the vibration test.

4.3.2.6 Resistance. The resistance of the indicator shall not be less than 40 ohms.

4.3.3 Sampling plan B tests. Testing shall be accordance with MIL-I-7071 sampling plan B testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:20Z
Check the source to verify that this is the current version before use.

4.3.3.1 Cold junction compensator lag. The indicator shall be connected to a chromel-alumel thermocouple (8 ohms total external circuit resistance), immersed in an ice bath of 0°C, and a millivoltage of constant value applied in series with the indicator and thermocouple to provide reading of 700°C. The indicator shall then be subjected to the conditions of the "Cold junction temperature lag" test specified in MIL-I-7071 for increasing and decreasing temperatures. The indication of the indicator throughout the test shall at no time vary by more than 30°C from the indication obtained at room temperature.

4.3.3.2 Extreme low temperature exposure. The scale errors at low temperature, as specified in the "Extreme low temperature exposure" test of MIL-I-7071, shall not exceed the tolerances specified in Table I. The scale errors at room temperature after low temperature exposure shall not exceed the tolerances specified in Table I.

4.3.3.3 High temperature exposure. The scale errors at room temperature after high temperature exposure, as specified in the "High temperature exposure" test of MIL-I-7071, shall not exceed the tolerances specified in Table I.

4.3.3.4 Thermal shock. No damage to the indicator shall result from the thermal shock test when subjected to the requirements the "Thermal shock" test of MIL-I-7071.

4.3.3.5 Salt fog. The scale errors at room temperature after salt fog exposure, as specified in MIL-STD-810, Method 509, shall not exceed the tolerances specified in Table I.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activities within the military service or defense agency or within the military service's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The thermocouple temperature indicator covered by this specification is intended to indicate the exhaust temperature of jet engines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:20Z
Check the source to verify that this is the current version before use.

b. If FAT required (see 3.3).
c. Packaging requirements (see 5.1).

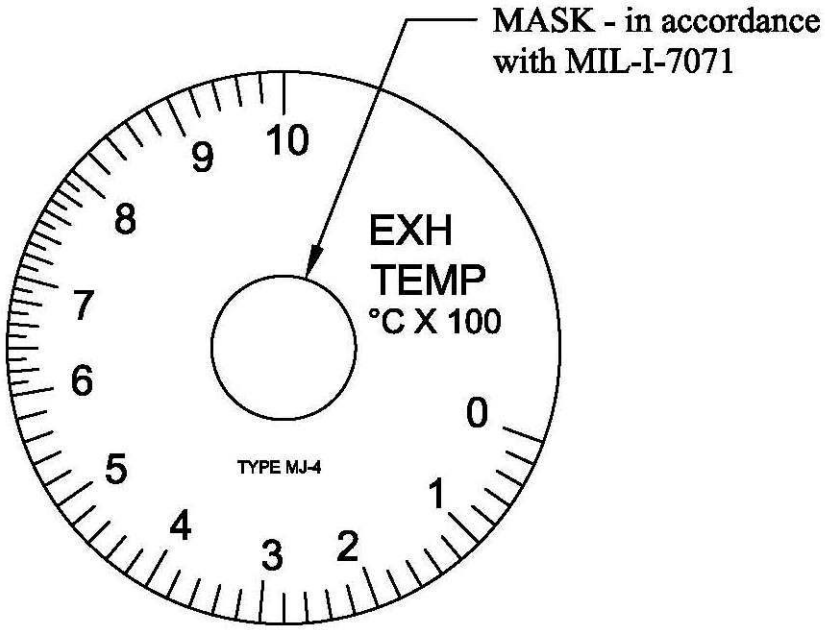
6.3 Bezel. The temperature indicator covered by this specification can be used on instrument panels in accordance with MS33639 by the application of an instrument mounting clamp in accordance with NASM6818.

6.4 Subject term (key word) listing.

air
aircraft
alumel
chromel
jet

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

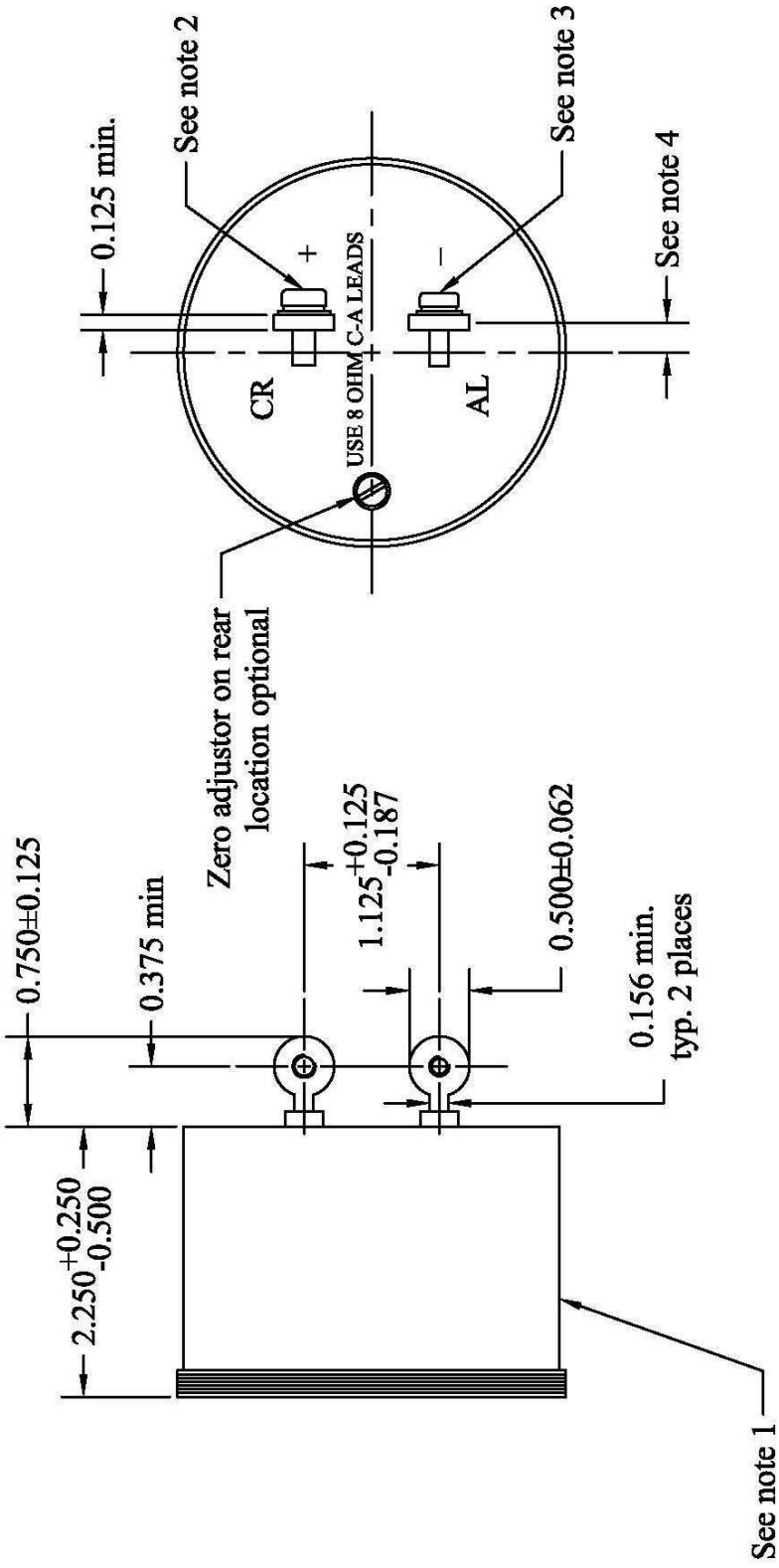
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:20Z
Check the source to verify that this is the current version before use.



Marking	Height or Length ±0.005 inches	Width of Line or Graduation ±0.005 inches	Material or Finish
Numerals	0.156	0.020	Fluorescent Luminescent Material
100°C Graduations	0.219	0.020	
20°C Graduations	0.156	0.010	
10°C Graduations	0.094	0.010	
Lettering			
"EXH TEMP"	0.156	0.020	
"°C x 100"	0.125	0.010	
Visible Portion of Pointer	---	---	
Type No.	0.062	0.010	Dull Black
Background of Dial	---	---	

Figure 1. Dial.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:20Z
Check the source to verify that this is the current version before use.



NOTES:
1. Case in accordance with MS33639, 3.250 inch nominal size.
2. Plus terminal, #10-32 UNF - screw per NASM53215-55, lock washer per NASM35338-100, washer per NASM691-10T.
3. Minus terminal, #8-32 UNC - screw per NASM35214-42, lock washer per NASM35338-99, washer per NASM961-8T
4. Terminals may be located 0.500 max. on either side of the vertical centerline
5. All dimensions in inches.
6. Break sharp edges approx. 0.010 radius.

Figure 2. Case

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:20Z
Check the source to verify that this is the current version before use.

Custodians:
Air Force – 99

Preparing Activity:
DLA – GS1

(Project 6685-2011-008)

Review Activity:
Air Force – 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T03:20Z
Check the source to verify that this is the current version before use.